

1. What is cloud computing?
2. How are cloud services delivered to clients?
3. What advantage does cloud computing provide in terms of upfront investment?
4. What is the NIST definition of cloud computing?
5. What types of resources are included in cloud computing?
6. What does “on-demand network access” mean in cloud computing?
7. What are the key supporting technologies behind cloud computing?
8. How does virtualization contribute to cloud computing?
9. What is meant by economies of scale in cloud computing?
10. What types of distributed systems support cloud applications?
11. What is utility computing in the context of cloud computing?
12. What is the pay-per-use model in cloud services?
13. What characteristics define an ideal cloud platform?
14. What does scalability mean in cloud computing?
15. What is elasticity in cloud systems?
16. What is resource pooling in cloud computing?
17. How does dynamic resource allocation work?
18. What does self-managing service mean in cloud computing?
19. How are cloud services made highly available and accessible?
20. What is interoperability and portability in cloud platforms?
21. What is over-provisioning in traditional computing?
22. What is under-provisioning and its impact?
23. What problems arise due to incorrect resource estimation?
24. How does dynamic provisioning solve resource issues?

25. What is the typical average server utilization?
26. Why do users provision systems for peak load?
27. What factors influence peak workload variations?
28. What are the four cloud deployment models?
29. What defines a public cloud?
30. What are the characteristics of public cloud infrastructure?
31. What is a private cloud?
32. What are the key features of private cloud systems?
33. What is a community cloud?
34. What is a hybrid cloud?
35. What are IaaS, PaaS, and SaaS?
36. What resources are provided by IaaS?
37. What does PaaS abstract for developers?
38. What does SaaS provide to end users?
39. Who are typical users of SaaS, PaaS, and IaaS (as shown in the diagram on page 19)?
40. What is virtualization?
41. What is a hypervisor?
42. How does a hypervisor manage multiple operating systems?
43. What is the trap-and-emulate model?
44. How are system calls handled in virtualization?
45. How are hardware interrupts managed by a hypervisor?
46. What happens when privileged instructions are executed in a guest OS?
47. What is VM context switching?

48. What are the steps involved in VM context switching?
49. What are Type 1 and Type 2 hypervisors?
50. What is KVM and how does it work in Linux?
51. What modules are used in KVM virtualization?
52. What is the role of QEMU in KVM?
53. What features are provided by OpenStack Compute REST API?
54. Why is IaaS alone insufficient for elastic computing?
55. What is multi-tenancy?
56. How does multi-tenancy improve resource utilization?
57. What are AMIs in cloud computing?
58. What is Platform-as-a-Service (PaaS)?
59. What is the difference between open PaaS and closed PaaS?
60. What are the major obstacles to cloud computing adoption?